Network-Based Diversification of Stock Portfolios

Network Science: Project Presentation

João Gouveia, Gonçalo Rua

Instituto Superior Técnico

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Presentation Overview

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Financial markets display complex dynamics:

Introduction

Motivation

Financial markets display complex dynamics:

- There aren't many techniques that can capture such dynamics.

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- There aren't many techniques that can capture such dynamics.
- Network science is a promising framework to capture these intricasies.

Traditional approaches to investing:

Introduction

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Using network science to automatically build a portfolio:

- Community detection algorithms identify distinct market segments.

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Using network science to automatically build a portfolio:

- Community detection algorithms identify distinct market segments.
- Network metrics identify the most desirable assets within each segment.

Kitanovski et al. [1] automate portfolio construction using:

Introduction

Previous Work

Kitanovski et al. [1] automate portfolio construction using:

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Their approach was then tested on the S&P 500 index.

Building on this work, we:

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- Explore different correlation measures.
- Explore different community detection algorithms.
- Analyze our approach on a wider subset of the global financial markets.

Automated Portfolio Construction

Overview

Three steps:

Automated Portfolio Construction

Overview

Three steps:

- Compute relations between the daily returns of assets.

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- Identify market segments based on these relations.

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Three steps:

- Compute relations between the daily returns of assets.
- Identify market segments based on these relations.
- Build a portfolio by selecting assets from each segment.

Automated Portfolio Construction

Asset Relations

Computing asset relations:

Automated Portfolio Construction

Asset Relations

Computing asset relations:

- Spearman Rank Correlation Coefficient (SCC).

Automated Portfolio Construction

Asset Relations

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- Spearman Rank Correlation Coefficient (SCC).
- Pearson Correlation Coefficient (PCC).

Automated Portfolio Construction

Asset Relations

Computing asset relations:

- Spearman Rank Correlation Coefficient (SCC).
- Pearson Correlation Coefficient (PCC).
- Co-Occurrence measures.

Four distinct Co-Occurrence measures:

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- Co-Occurrence SCC using Louvain to identify the monthly communities.

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Automated Portfolio Construction

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Computing a Co-Occurrence measure:

Automated Portfolio Construction

Asset Relations

Computing a Co-Occurrence measure:

- Compute the monthly SCC/PCC relations between assets.

Automated Portfolio Construction

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Automated Portfolio Construction

Asset Relations

Computing a Co-Occurrence measure:

- Compute the monthly SCC/PCC relations between assets.
- Identify the market segments for each month using a community detection algorithm.
- For each pair of assets, compute the frequency with which they belong to the same community.

In short, we have 6 possibile measures of asset relations:

- SCC.
- PCC.
- Four Co-Occurrence measures.

Automated Portfolio Construction

Network Representation

We now have, for each pair of assets, a number reflecting how they are related.

Automated Portfolio Construction

Network Representation

We now have, for each pair of assets, a number reflecting how they are related.

Goal: Use these values to determine groups of assets that are closely related.

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Network Representation

Build a fully connected graph:

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Network Representation

Build a fully connected graph:

- One node for each available asset.

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Network Representation

Build a fully connected graph:

- One node for each available asset.
- Use the relation measures between assets as edge weigths.

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Network Representation

Problem: Most community detection algorithms are designed for sparse networks.

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Network Representation

Problem: Most community detection algorithms are designed for sparse networks.

- Our fully connected graph is not suitable for this.

Solution: Compute the *Minimum Spanning Tree* of our graph.

- Yields a graph amenable to community detection.

Automated Portfolio Construction

Community Detection

Identifying market segments:

Automated Portfolio Construction

Community Detection

Identifying market segments:

- Louvain Method.

Automated Portfolio Construction

Community Detection

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Automated Portfolio Construction

Asset Selection

To build the final portfolio:

Automated Portfolio Construction

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To build the final portfolio:

- Select 25 assets proportionally from each community to maximize diversification.

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Asset Selection

To build the final portfolio:

- Select 25 assets proportionally from each community to maximize diversification.
- Use network metrics to rank assets within each community.

Automated Portfolio Construction

Asset Selection

Selection metrics:

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Asset Selection

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- Principal Component Analysis (PCA): asset's exposure to dominant market-wide risk factor.

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- Principal Component Analysis (PCA): asset's exposure to dominant market-wide risk factor.
- Degree Centrality (CD): total strength of an asset within its community.
- Closeness Centrality (CC): average shortest path from an asset to all others in the network.

Selection Metric Ranges:

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- Med: select assets around the median value.
- Max: select assets with the highest metric scores - typically more central.

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Rationale:

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Asset Selection

Rationale:

- Peripheral assets may help reduce volatility.

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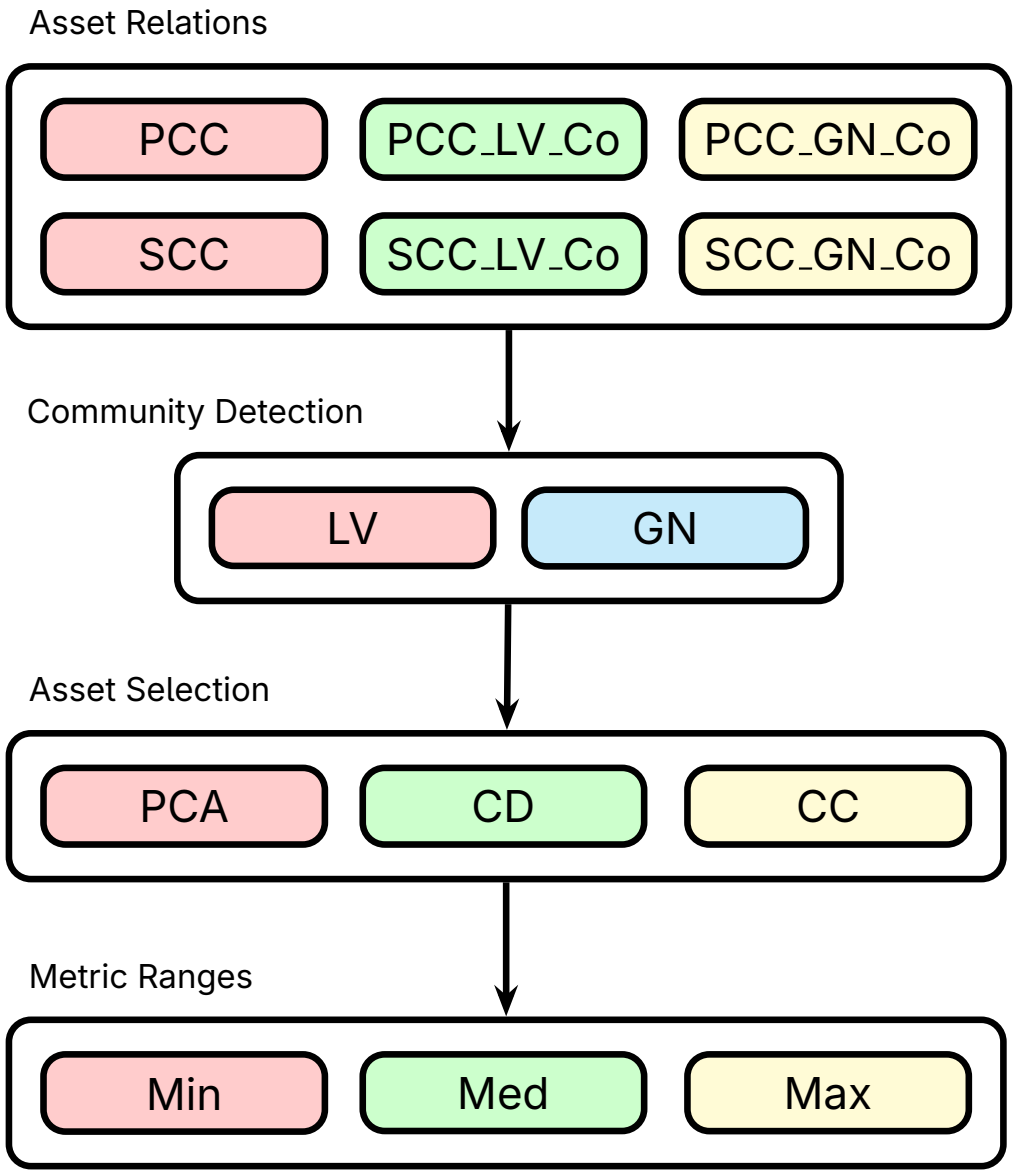
Asset Selection

Rationale:

- Peripheral assets may help reduce volatility.
- Central assets may capture core market movements.

Automated Portfolio Construction

Summary



Results

Performance

We evaluated 108 strategies across:

Results

Performance

We evaluated 108 strategies across:

- **S&P 500 Index**.

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We evaluated 108 strategies across:

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- **FTSE 100 Index**.

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Best-performing strategies:

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Best-performing strategies:

- S&P 500: PCC + Louvain + Min PCA.

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- **S&P 500 Index**.
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Best-performing strategies:

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- FTSE 100: SCC_GN_Co + Girvan-Newman + Min CD.

Results

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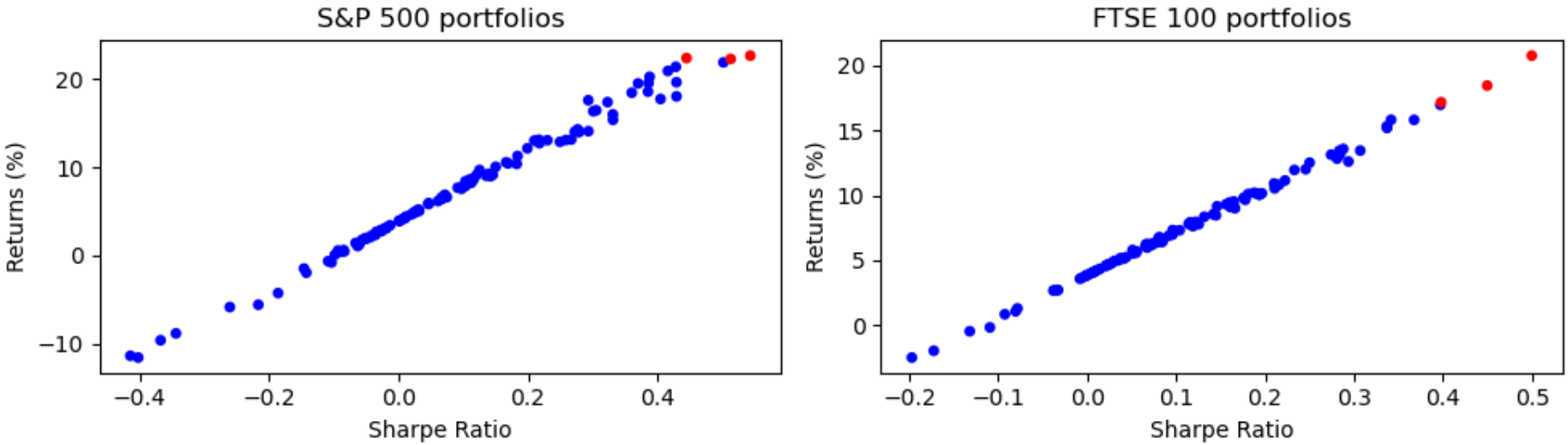


Figure: Returns

- Sharpe ratio: compares portfolio returns with its risk

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Performance

- The best strategies use either minimal PCA or minimal CD scores.
- Minimal PCA score reflects the lowest exposure to the market's dominant factor.
- Minimal CD scores reflect the weakest connection to immediate neighbors.
- This confirms the principle of favoring **peripheral assets** to reduce portfolio volatility.

Results

Robustness

Goal: Assess how strategies perform across markets.

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- Measured the absolute difference in performance between S&P 500 and FTSE 100.

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- PCC + GN + Max CD.
- SCC + GN + Median CD.

Results

Robustness

Insights:

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- Robust strategies consistently used Girvan–Newman for community detection.

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Robustness

Insights:

- Robust strategies consistently used Girvan–Newman for community detection.
- Degree Centrality was the most robust selection metric.
- Mixing Louvain for co-occurrence and GN for segmentation appears promising.

Results

Difference in results from the previous work

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- Our results differ greatly from the work of Kitanovski et al.

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- we found the opposite to be true.
- Explained by their usage of market data from the COVID-19 pandemic and the Ukraine invasion?

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- Performance highly sensitive to both the specific market and the training period selected.

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Conclusions

- Performance highly sensitive to both the specific market and the training period selected.
- Strategies are unable to fully capture the complex dynamics of financial markets.
- Designing a method that generalizes well across scenarios remains a significant challenge.

Bibliography

[1] Dimitar Kitanovski et al. “Network-based diversification of stock and cryptocurrency portfolios”. In: *Applied Network Science* 10.20 (2025), pp. 1–46. DOI: 10.1007/s41109-025-00708-9.